

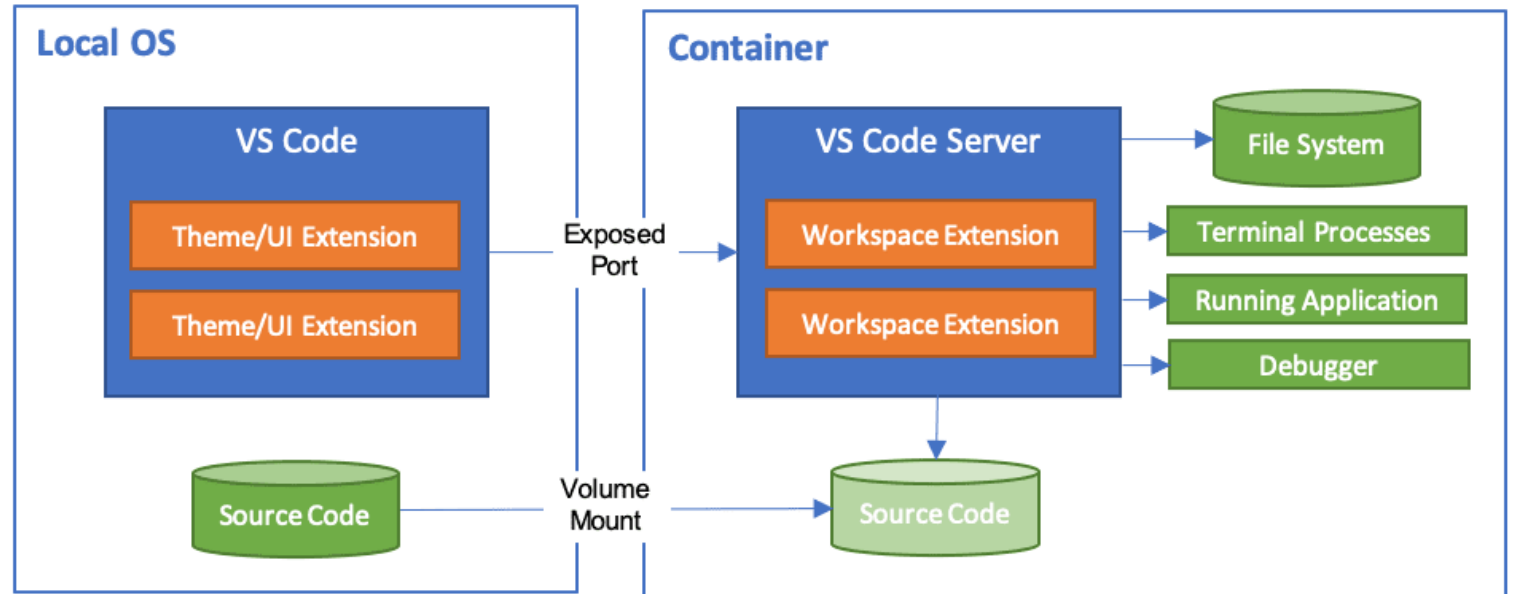
Docker Installation Instructions

Requirements

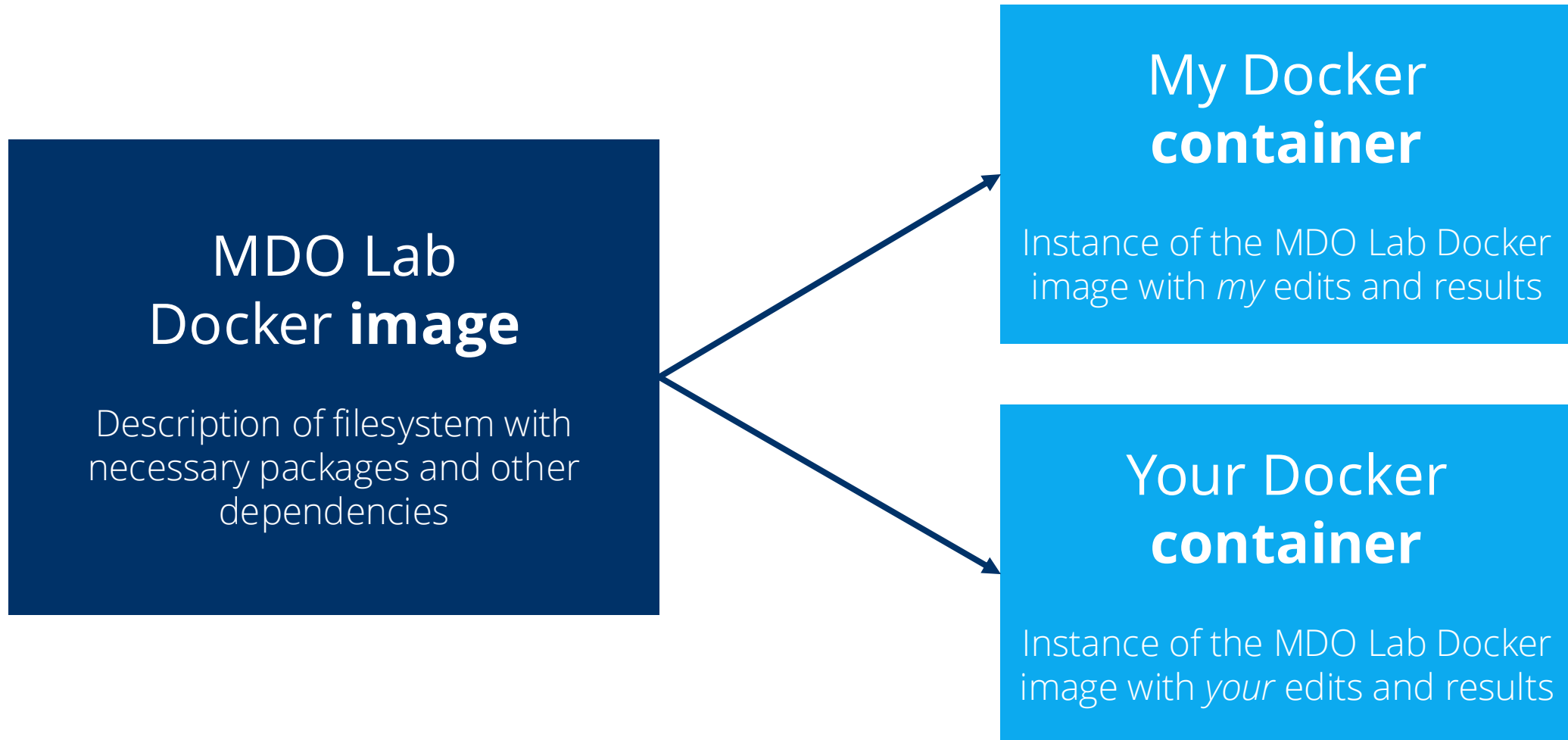
- Computer with admin access
- macOS, Windows 10/11, or Linux distribution
- 64-bit processor and operating system
- 4GB of RAM

Use a Docker container to run the codes

- Avoid manually compiling and installing all packages and their dependencies
- Eliminate operating system differences
- Separate filesystem from your computer



Docker terminology



Setup steps

1. Install Docker
2. Install VS Code
3. Download the MDO Lab Docker image
4. Create a container (an instance of the image you can use)

Step 1a of 4: Install Docker

Windows

*Use WSL 2 backend(see instructions)

<https://docs.docker.com/desktop/setup/install/windows-install/>

macOS

<https://docs.docker.com/desktop/setup/install/mac-install/>

Linux

<https://docs.docker.com/desktop/setup/install/linux/>

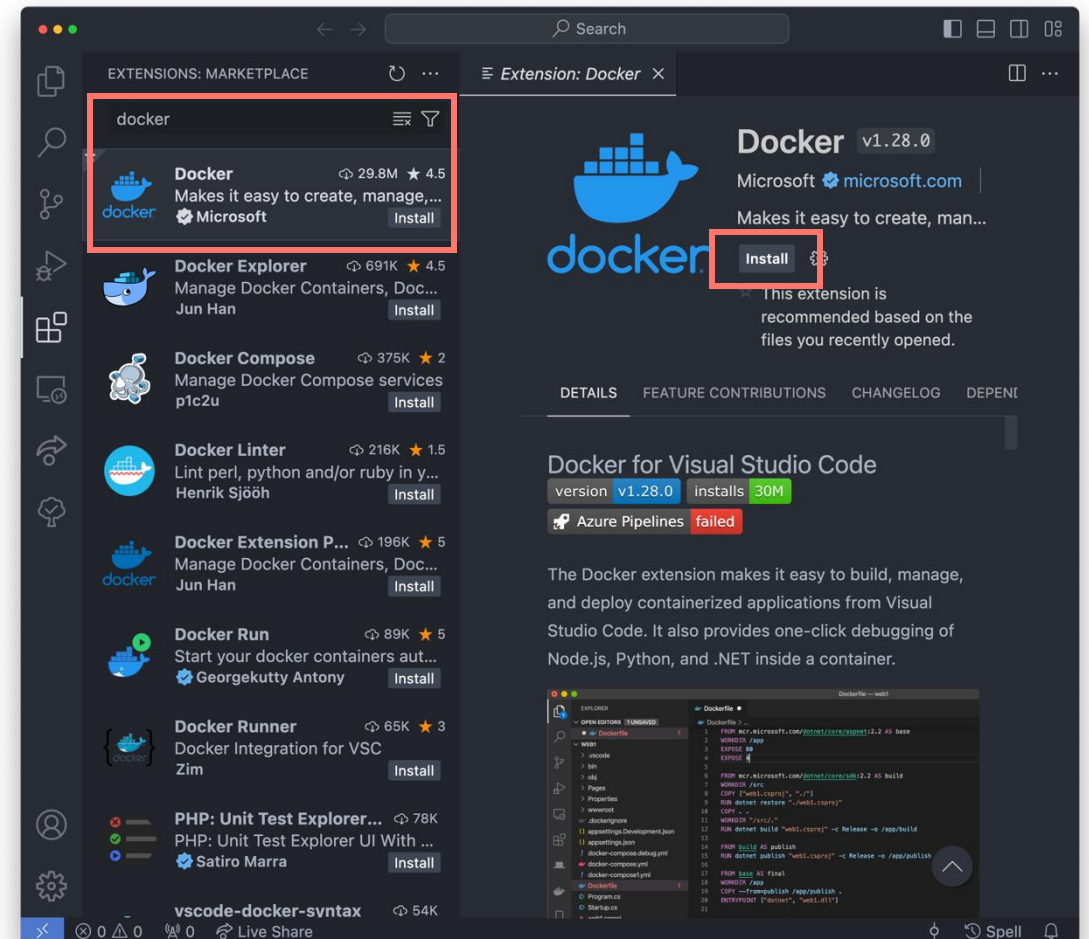
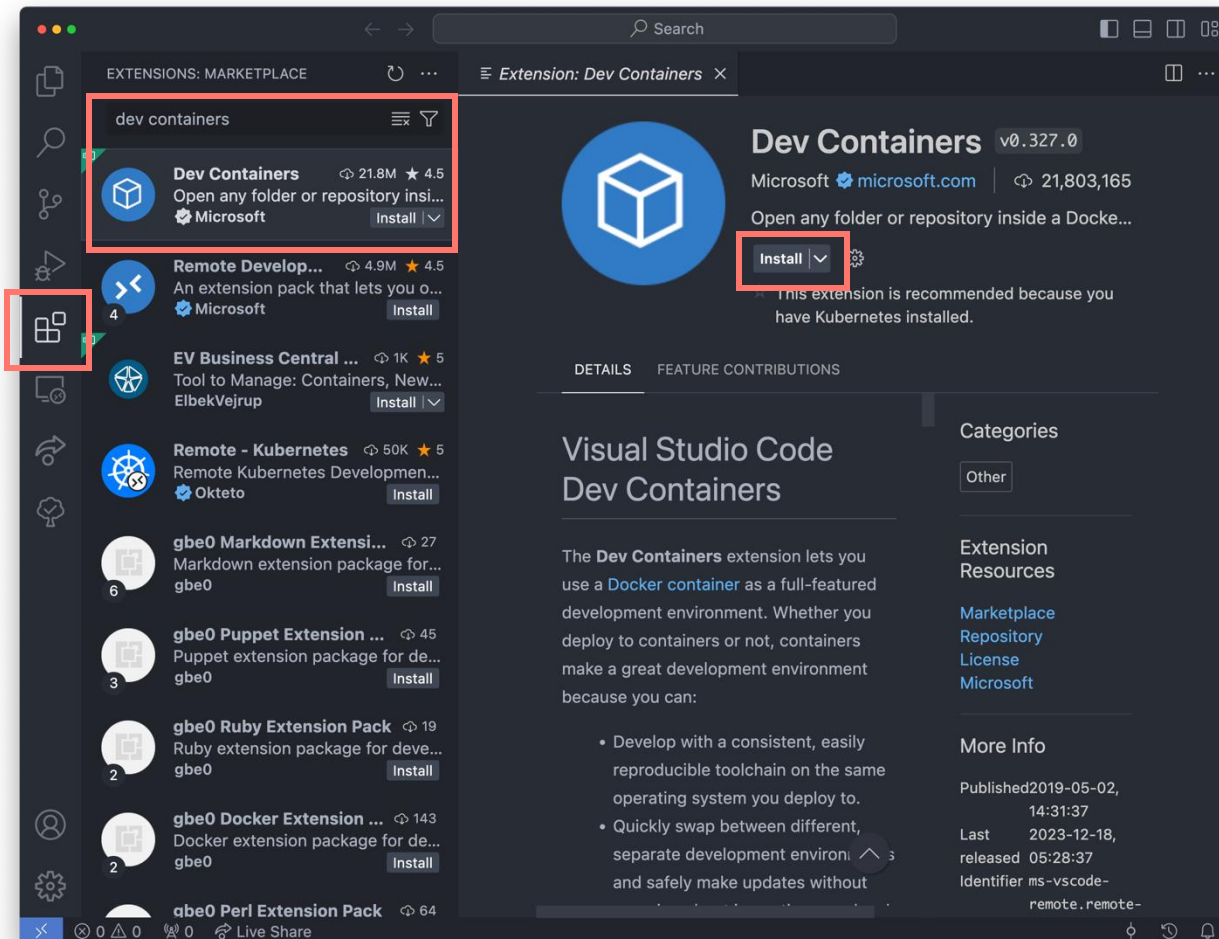
Step 1b of 4: Check Docker install

In a terminal if on Mac or Linux
or Powershell on Windows:

```
docker run hello-world
```

Step 2 of 4: Install VS Code and extensions

code.visualstudio.com/download



Step 3 of 4: Pull MDO Lab Docker image

In a terminal if on Mac or Linux
or Powershell if on Windows:

```
docker pull mdolab/public:u22-gcc-mpi-latest
```

Step 4a of 4: Create container

In a terminal if on Mac or Linux or Powershell if on Windows:

Create directory on your host OS to pass files into/out of the container

```
mkdir ~/ae740_docker
```

```
cd ~/ae740_docker
```

```
pwd
```

 (replace hostdir in the command below with the output of this command)

Create the Docker container and mount the local directory to it

```
docker run -it --name ae740 --mount  
"type=bind,src=hostdir,target=/home/mdolabuser/mount"  
mdolab/public:u22-gcc-mpi-latest /bin/bash
```

```
exit
```

Step 4b of 4: Check container

In a terminal if on Mac or Linux or Powershell if on Windows:

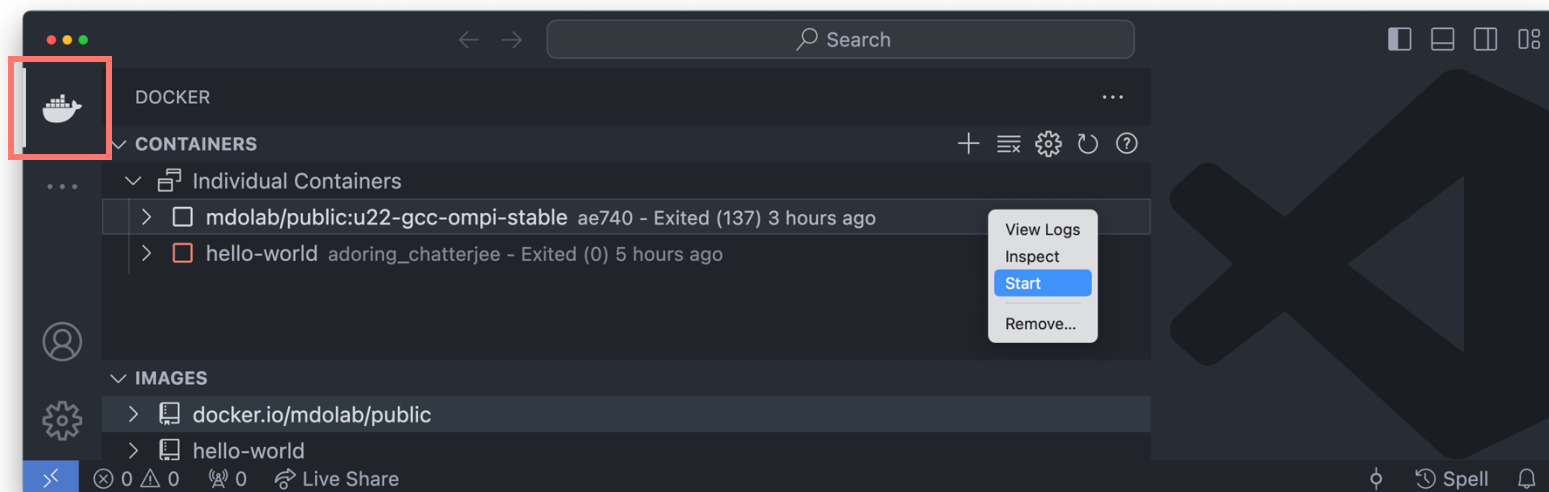
```
docker container ls -a
```

Should print out text with a line like the highlighted one

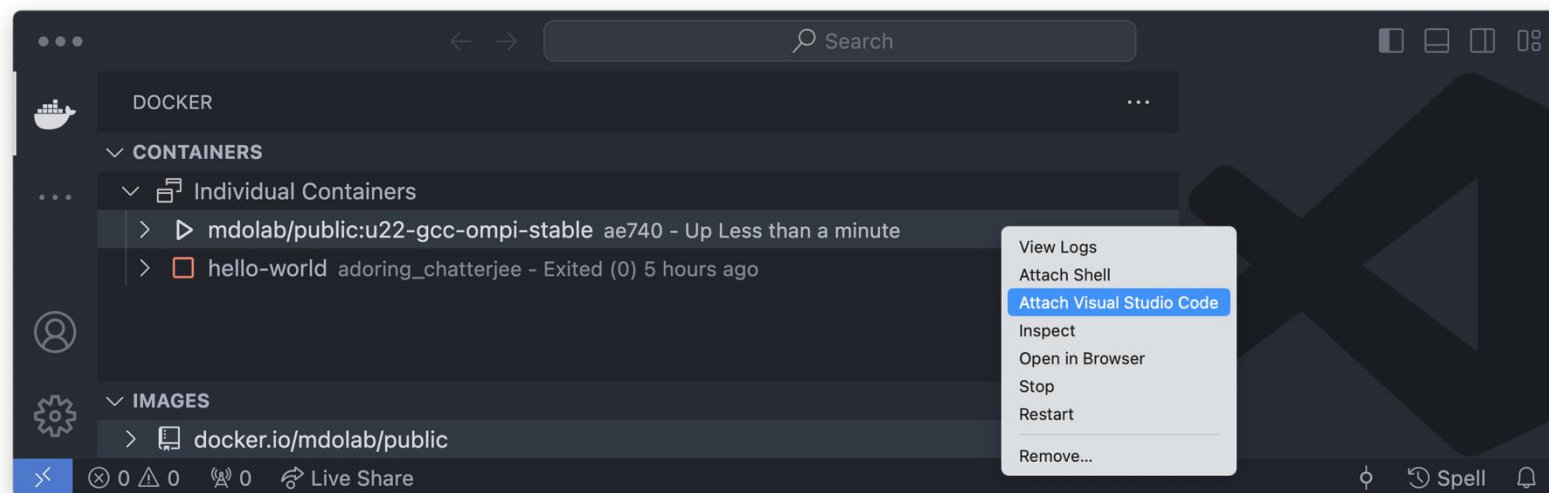
CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
hjd6jsr65421	mdolab/public:u22-gcc-mpi-stable	"/bin/bash"	7 seconds ago	Exited (0) 4 seconds ago		ae740
s6a53j3ah5a5	hello-world	"/hello"	20 minutes ago	Exited (0) 5 minutes ago		adoring_Chatterjee

Launching Docker from VS Code

Start the container
(right click)

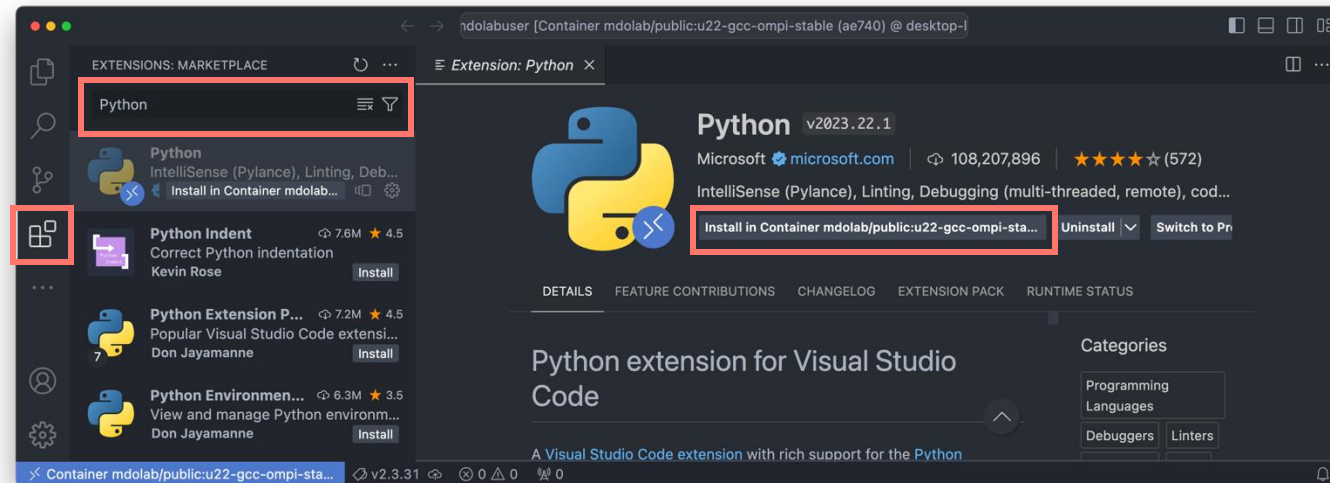


Attach a VS Code
window to it



Final things to install into the container

1. Terminal > New terminal (might need to type `/bin/bash` after)
2. `sudo apt-get update`
 - Password is "temppass"
3. `sudo apt-get install -y ffmpeg xfoil`
4. VS Code's Python extension:



When you're done (don't do this now)

